

Agile Team Project

Week 5: Part 3
CI/CD - Masterclass

This week's masterclass:

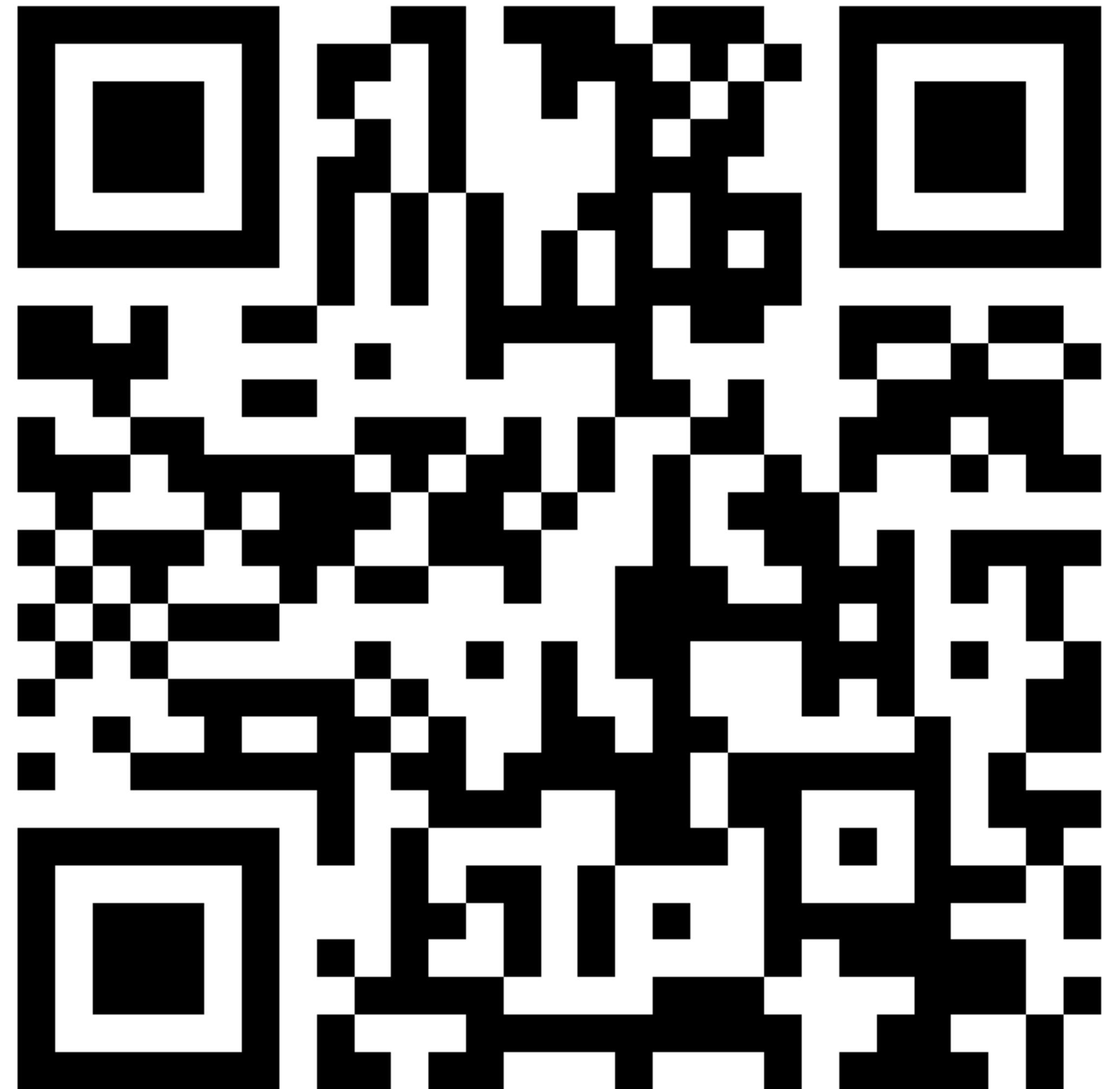
Get CI/CD working

0) get updated CI/CD workflow + scripts from team00 repo

- 1) get an Azure Cloud Virtual Machine: ip + username + ssh private key
- 2) request GitHub repo admin access
- 3) request DNS entries `teamNN.bham.team` and `teamNN.dev.bham.team`
- 4) install docker on Cloud Virtual Machine
- 5) install GitHub runner on Cloud Virtual Machine
- 6) commit to main so `publish-docker:` and `deploy-dev:` will run
- 7) add GitHub secrets:
 - Cloud Virtual Machine: ip address + username + ssh private key
- 8) create a tagged commit so `deploy-dev:` will run
- 9) get real `https://` SSL certificates - change ACME in `main.yml`
- 10) get custom url working - edit Caddyfile

changes from feedback form:

- Given tech stack
- Less busy work
- More tech masterclasses
- (JDL, Angular, SpringBoot, Deployment)
- Hybrid-groups
- Ranking
- More tech TA support
- Provided tech stack+VM+pipelines



Get updated CI/CD workflow + scripts from team00 repo v2

```
.github/workflows/main.yml  
src/main/docker/Caddyfile  
src/main/docker/dev.yml  
src/main/docker/prd.yml  
src/main/docker/install-app.sh
```

<https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/.github/workflows/main.yml>

<https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/Caddyfile>

<https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/dev.yml>

<https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/prd.yml>

<https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/install-app.sh>

get latest version of the CI files using the terminal

- 1) cd to the root folder of your repo
- 2) Run the following commands:



```
curl -fsSL https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/.github/workflows/main.yml -o .github/workflows/main.yml
curl -fsSL https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/Caddyfile -o src/main/docker/Caddyfile
curl -fsSL https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/dev.yml -o src/main/docker/dev.yml
curl -fsSL https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/prd.yml -o src/main/docker/prd.yml
curl -fsSL https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/install-app.sh -o src/main/docker/install-app.sh
```

<https://code-pad.me/winter-oklahoma-table-connecticut>

After getting the new files: commit these files to git

Create cloud VM in Azure

- <https://go.microsoft.com/fwlink/?linkid=2062405&clcid=0x809>
- provides \$100 (£80) credit - no credit card needed - can be renewed yearly for students. Takes 5 minutes. Looks like this:

The image shows two overlapping screenshots from the Microsoft Azure portal. The background screenshot is the 'Academic Verification' page, which has a dark header with the Microsoft Azure logo. The page title is 'Academic Verification' with an upward arrow icon. Below the title, there is instructional text: 'Start by entering your name as per the school records. Select your school's country and enter your school's name. Enter your date of birth as per the school records.' There are three input fields: 'First name' with the value 'Madasar', 'Last name' with the value 'Shah', and 'Country' with a dropdown menu showing 'United Kingdom'. The foreground screenshot is the 'Education | Overview' page, which has a blue header with the Microsoft Azure logo. The page title is 'Education | Overview' with a graduation cap icon and a star icon. Below the title, there is a navigation bar with 'Overview' selected, 'Learning resources', and 'Need help?'. On the right side, there is a 'Student offer details' section with three items: 'Available credits' showing '£58 out of £80', 'Days until credit expires' showing '334' and 'Expires on 14/01/2026', and 'February costs' showing '£9.94'.

Microsoft Azure

Academic Verification

Start by entering your name as per the school records. Select your school's country and enter your school's name. Enter your date of birth as per the school records.

First name

Madasar

Last name

Shah

Country

United Kingdom

Microsoft Azure

Home >

Education | Overview

Get started Overview

Overview

> Learning resources

> Need help?

Student offer details

Available credits

£58 out of £80

Days until credit expires

334

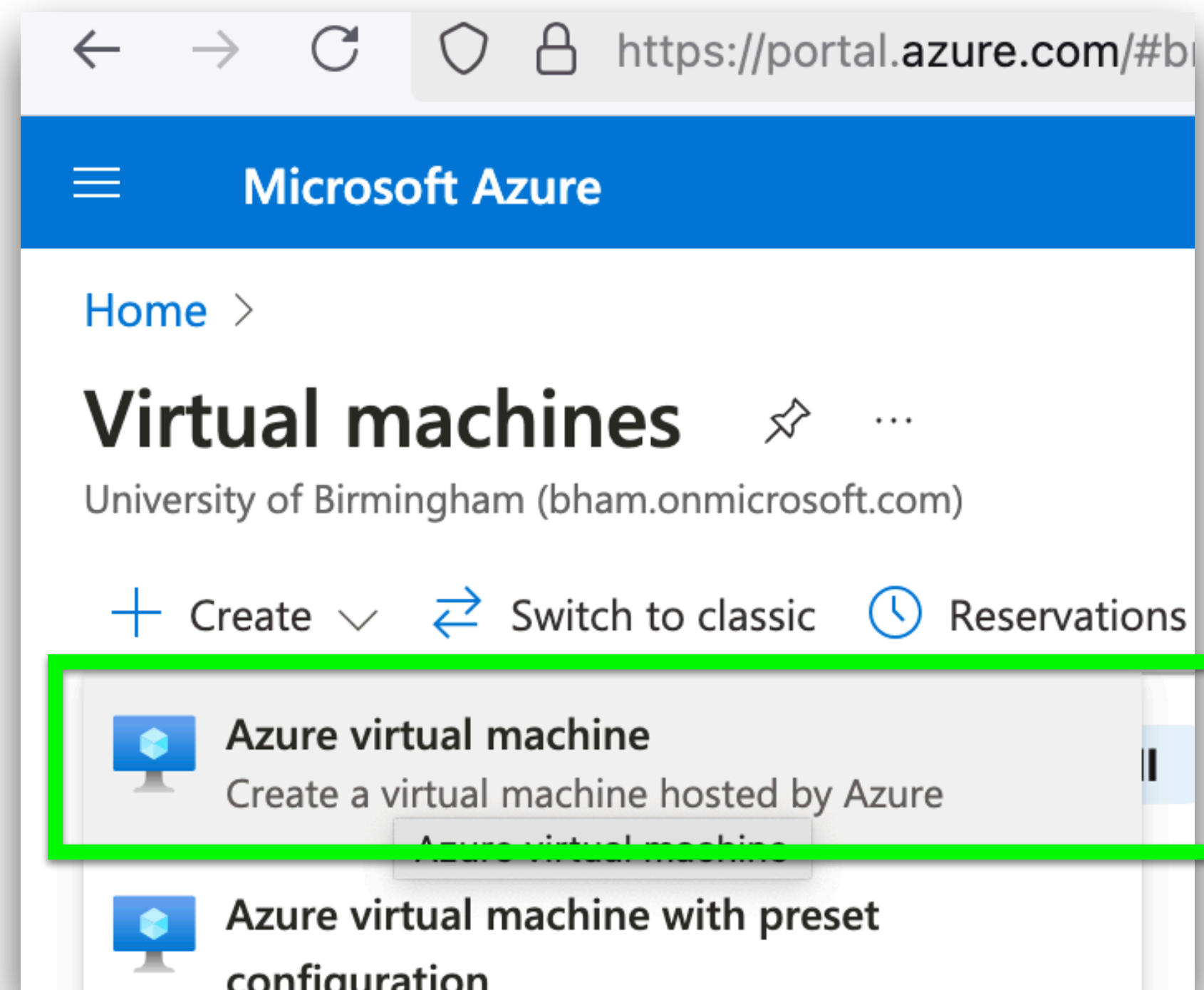
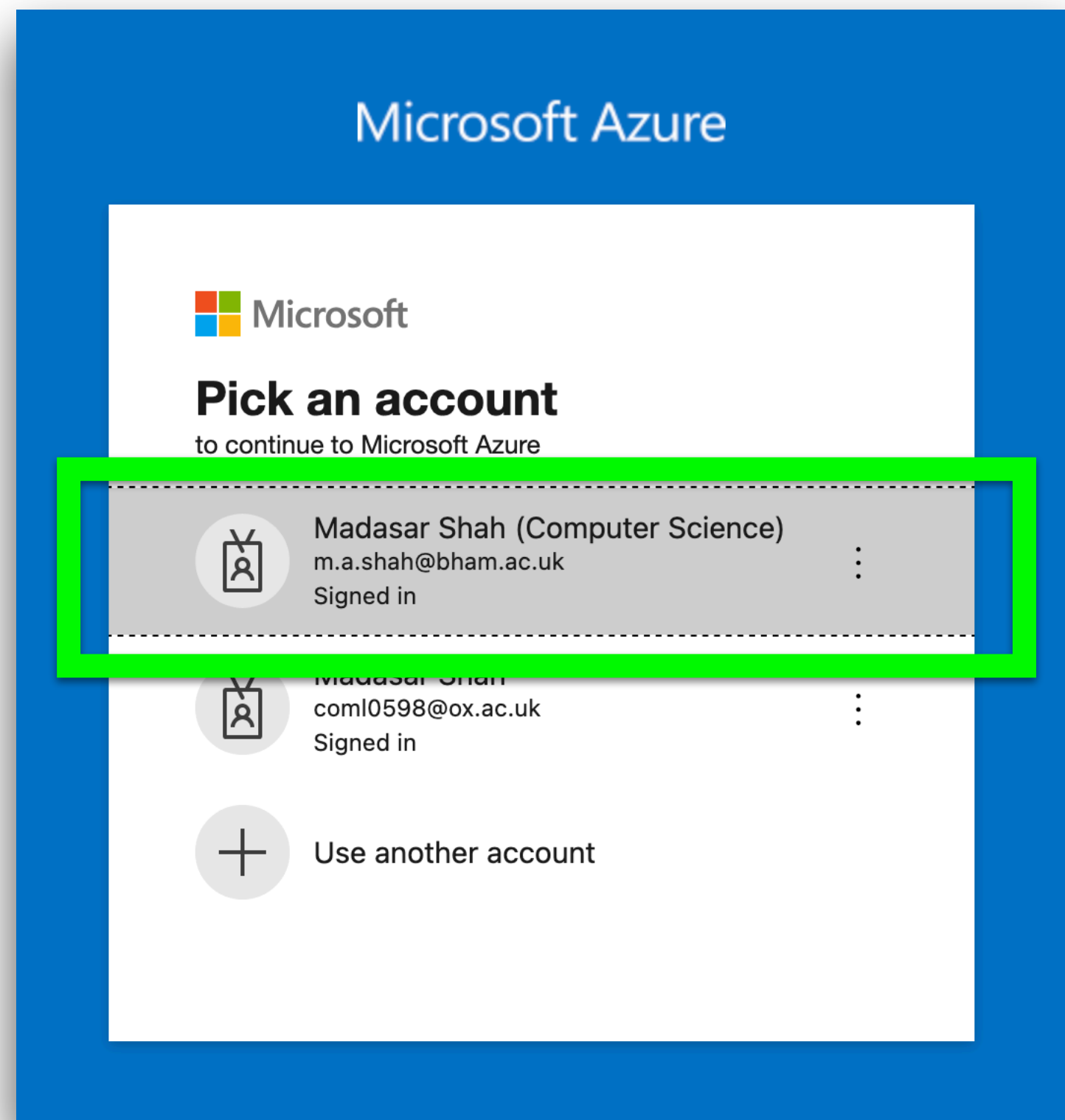
Expires on 14/01/2026

February costs

£9.94

Create cloud VM in Azure

- <https://portal.azure.com/#browse/Microsoft.Compute%2FVirtualMachines>



Create cloud VM in Azure

Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students
Resource group * (New) team00_group
[Create new](#)

Instance details

Virtual machine name * team00
Region * (Asia Pacific) Central India
Availability options No infrastructure redundancy required

Security type Trusted launch virtual machines
[Configure security features](#)

Trusted launch virtual machine is required when using 1P Gallery images.

Image * Ubuntu Server 24.04 LTS - x64 Gen2
[See all images](#) | [Configure VM generation](#)

VM architecture Arm64
x64

Run with Azure Spot discount

Size * Standard_B2als_v2 - 2 vcpus, 4 GiB memory (US\$17.96/month)
[See all sizes](#)

Administrator account

Authentication type

SSH public key
Password

Azure now automatically generates an SSH key pair for you and allows you to store it for future use. It is a fast, simple, and secure way to connect to your virtual machine.

Username *

ubuntu

SSH public key source

Generate new key pair

SSH Key Type

RSA SSH Format
Ed25519 SSH Format

Ed25519 provides a fixed security level of no more than 128 bits for 256-bit key, while RSA could offer better security with keys longer than 3072 bits.

Key pair name *

team00_key

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports *

None
Allow selected ports

Select inbound ports *

HTTP (80), HTTPS (443), SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

cloud VM in Azure

Generate new key pair

i An SSH key pair contains both a public key and a private key. **Azure doesn't store the private key.** After the SSH key resource is created, you won't be able to download the private key again. [Learn more](#)

Download private key and create resource

Return to create a virtual machine



team00_key.pem

← → ↻ 🔒 https://portal.azure.com/#home

Microsoft Azure Search resources, services, and docs (G+)

Azure services

- Create a resource
- Virtual machines
- Virtual networks
- Cost Management...

Resources

Recent Favorite

team00

team00_group

See all

Microsoft Azure Search resources, services, and docs (G+)

Home > **team00** Virtual machine

Search

Overview

- Activity log
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- Tags
- Diagnose and solve problems
- Connect
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- Availability + scale
- Security
- Backup + disaster recovery
- Operations
- Monitoring
- Automation
- Help

team00 virtual machine agent status is not ready. Troubleshoot the issue →

Help me copy this VM in any region

Connect Start Restart Stop Hibernate Capture Delete Refresh Open in mobile Feedback

Essentials

Resource group (move)	: team00_group	Operating system	: Linux
Status	: Running	Size	: Standard D4s v4 4 GiB memory
Location	: Central India	Public IP address	: 74.225.135.239
Subscription (move)	: Azure for Students	Virtual network/subnet	: team00-vnet/default
Subscription ID	: 19a564c1-941d-4497-8dee-b95a79a23be7	DNS name	: Not configured
		Health state	: -
		Time created	: 16/02/2025, 14:09 UTC

Tags (edit) : Add tags

Properties

Virtual machine		Networking	
Computer name	: team00	Public IP address	: 74.225.135.239 (Network interface)
Operating system	: Linux	Public IP address (IPv6)	: -

SSH into the Azure VM

Need:

```
~/Downloads/team00_key.pem - key file from Azure  
ubuntu - admin username of VM we set in Azure  
74.225.135.239 - VM IP address from Azure
```

```
cat ~/Downloads/team99_key.pem | pbcopy
```

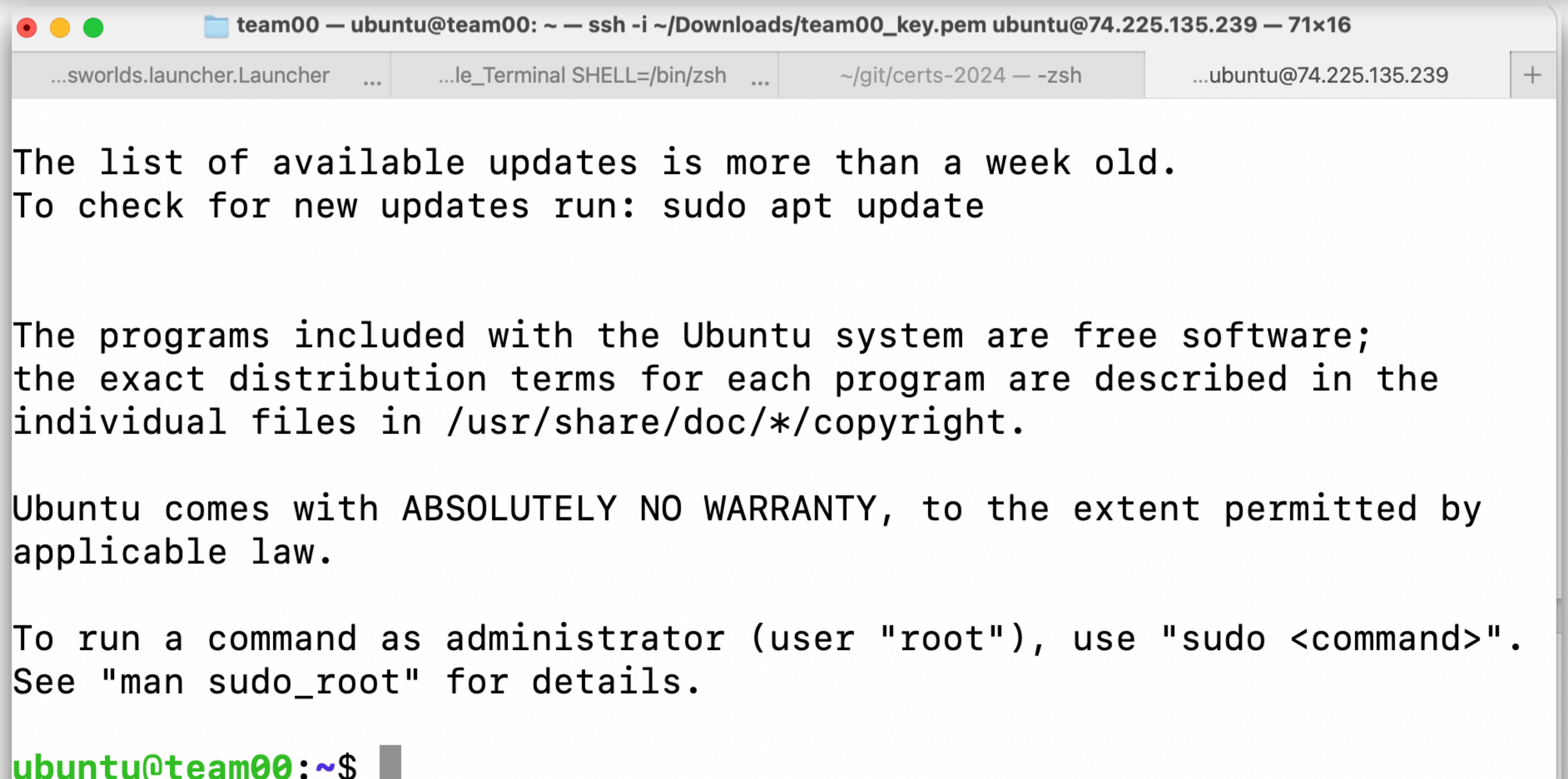
Run:

```
chmod go-rx ~/Downloads/team00_key.pem  
ssh -i ~/Downloads/team00_key.pem ubuntu@74.225.135.239
```

Keyfile permissions must be:

```
shah@mars emp % ls -la ~/Downloads/team00_key.pem  
-rw-----@ 1 shah  staff  2498 16 Feb 14:09 /Users/shah/Downloads/team00_key.pem
```

To get shell access to the cloud VM:



The image shows a terminal window titled "team00 — ubuntu@team00: ~ — ssh -i ~/Downloads/team00_key.pem ubuntu@74.225.135.239 — 71x16". The terminal output displays the Ubuntu login banner, which includes information about updates, software licenses, and warranty. The prompt "ubuntu@team00:~\$" is visible at the bottom.

```
team00 — ubuntu@team00: ~ — ssh -i ~/Downloads/team00_key.pem ubuntu@74.225.135.239 — 71x16  
...sworlds.launcher.Launcher ... ...le_Terminal SHELL=/bin/zsh ... ~/git/certs-2024 — -zsh ...ubuntu@74.225.135.239 +  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@team00:~$
```

Get admin access to repository

<https://forms.office.com/e/qWl4MhIusn>

Request admin rights for the team repository

We collect names and uni email addresses automatically.

Please submit this form with minuted agreement from your team - record this agreement in writing in the team diary.

We will grant admin rights to one team mate to add runners, add github se

Do not delete your repository - we cannot recover the repository if

Do not allow rebase merges.

Do not use the shared runners from the github organisation.

Only submit this form if the GitHub username in question 2 belongs to you.

If the member with admin rights needs to change, please submit the form :

Hi, Madasar. When you submit this form, the owner will see your name and

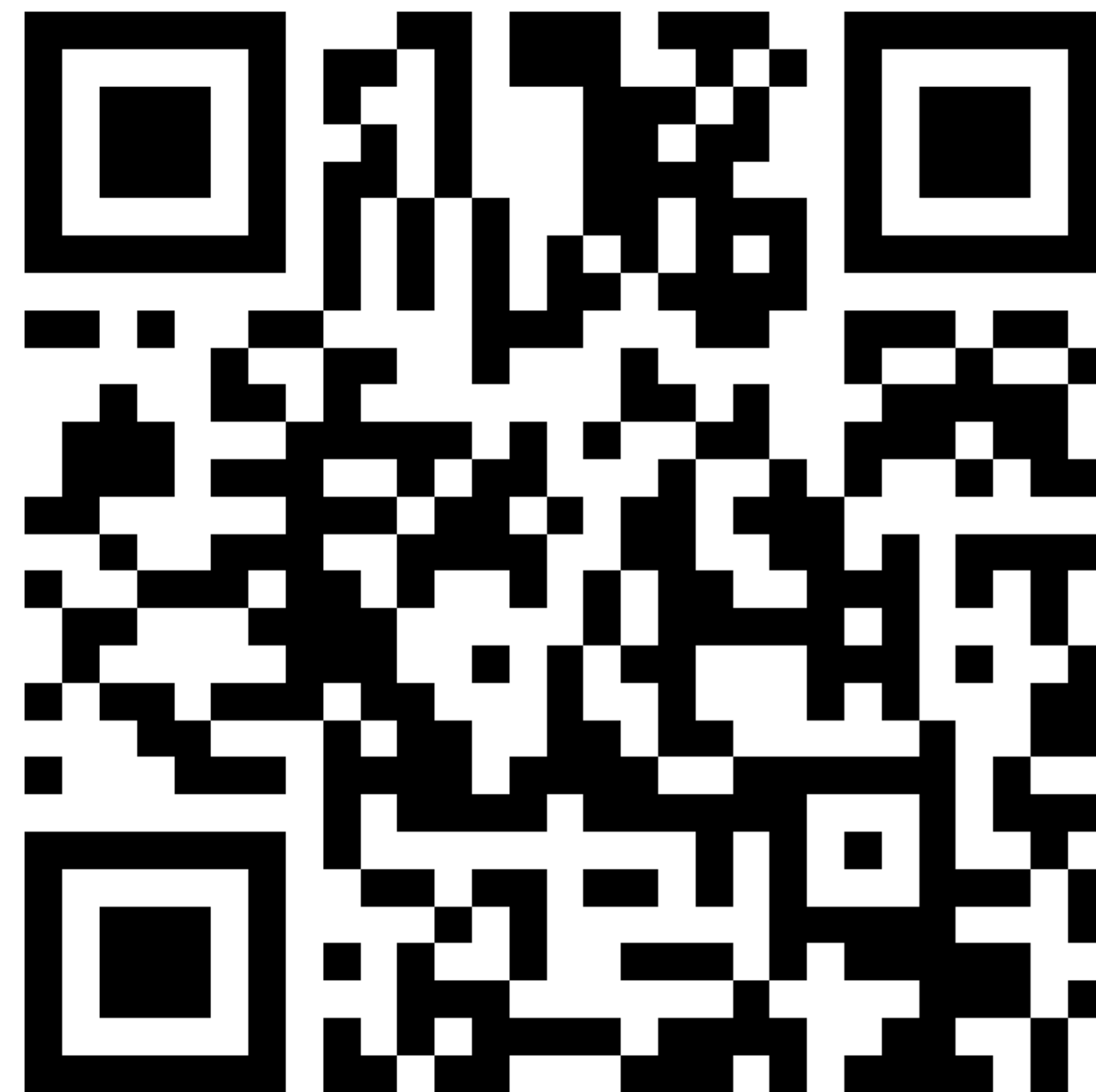
* Required

1. Which team are you are in *

team01

2. What is your GitHub username *

madasar



← → ↻ 🔒 https://github.com/uob-team-project/team00/settings

☰ GitHub uob-team-project / team00

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⚙️ General

Access

- 👤 Collaborators and teams
- 🔒 Moderation options

Code and automation

- 🌿 Branches
- 🏷️ Tags
- 📄 Rules
- 🕒 Actions
- 🔗 Webhooks
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General

Repository name

team00 Rename

☒ **Template repository**

Template repositories let users generate new repositories with the same [repositories](#).

☐ **Require contributors to sign off on web-based commits**

Enabling this setting will require contributors to sign off on commits or contributors to affirm that their commit complies with the repository's [Learn more about signing off on commits](#).

Default branch

The default branch is considered the "base" branch in your repository. Commits are automatically made, unless you specify a different branch.

main ✎

Request DNS entries for cloud VM

<https://forms.office.com/e/C0kaiRMuDD>

DNS Entry Request

Use this form to request a DNS entry for your Virtual Machine. The team will be automatically be allocated a URL in the form [team00.dev.bham.team](#) and [team00.bham.team](#) that will point to the IP address you give here. **These URLs will be used to assess the teams' apps in M2.**

We will allow the team to request a custom domain e.g. [appname.bham.team](#) - the team will be responsible for setting up https and forwarding. [team00.bham.team](#) and [team00.dev.bham.team](#) domains must always work.

Please submit this form with minuted agreement from your team.

By default we will add a DNS A record for the IP virtual machine.

If the ip address or custom domain changes please submit the form again.

Hi, Madasar. When you submit this form, the owner will see your request.

* Required

1. Which team are you are in *

team01

2. What is the IPv4 address of your Virtual Machine *

in the format x.x.x.x - where x = 0-256, no spaces or text

74.225.135.239



```
shah@mars team00 % nslookup team00.bham.team
Server:                8.8.8.8
Address:                8.8.8.8#53
```

Non-authoritative answer:

```
Name:    team00.bham.team
Address: 74.225.135.239
```

```
shah@mars team00 % nslookup team00.dev.bham.team
Server:                8.8.8.8
Address:                8.8.8.8#53
```

Non-authoritative answer:

```
Name:    team00.dev.bham.team
Address: 74.225.135.239
```

Install Docker on Azure cloud VM

```
curl -fsSL https://get.docker.com -o get-docker.sh && sh get-docker.sh
sudo usermod -a -G docker $USER
newgrp docker
docker ps
```

```
ubuntu@team00:~$ curl -fsSL https://get.docker.com -
# Executing docker install script, commit: 4c94a5699
+ sudo -E sh -c apt-get -qq update >/dev/null
+ sudo -E sh -c DEBIAN_FRONTEND=noninteractive apt-get
dev/null
+ sudo -E sh -c install -m 0755 -d /etc/apt/keyrings
+ sudo -E sh -c curl -fsSL "https://download.docker.
s/docker.asc
+ sudo -E sh -c chmod a+r /etc/apt/keyrings/docker.a
+ sudo -E sh -c echo "deb [arch=amd64 signed-by=/etc
d.docker.com/linux/ubuntu noble stable" > /etc/apt/s
+ sudo -E sh -c apt-get -qq update >/dev/null
+ sudo -E sh -c DEBIAN_FRONTEND=noninteractive apt-get
containerd.io docker-compose-plugin docker-ce-rootl
l
Scanning processes...
Scanning linux images...
+ sudo -E sh -c docker version
Client: Docker Engine - Community
Version:      27.5.1
API version:  1.47
```

```
team00 — ubuntu@team00: ~ — ssh -i ~/Downloads/team00_key.pem ubuntu@74.225.135.239 — 93x24
...s.plexus.classworlds.launcher.Launcher ... ...GRAM=Apple_Terminal SHELL=/bin/zsh ... ~/git/certs-2024 — -zsh ...00_key.pem ubuntu@74.225.135.239 +

dockerd-rootless-setup tool.sh install

Visit https://docs.docker.com/go/rootless/ to learn about rootless mode.

To run the Docker daemon as a fully privileged service, but granting non-root
users access, refer to https://docs.docker.com/go/daemon-access/

WARNING: Access to the remote API on a privileged Docker daemon is equivalent
to root access on the host. Refer to the 'Docker daemon attack surface'
documentation for details: https://docs.docker.com/go/attack-surface/
=====
ubuntu@team00:~$ sudo usermod -a -G docker $USER
ubuntu@team00:~$ docker ps
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/dock
er.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.47/containers/json": dial unix /var/run/do
cker.sock: connect: permission denied
ubuntu@team00:~$ newgrp docker
ubuntu@team00:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@team00:~$
```

Install GitHub Action Runner on Azure Cloud VM

Get repo admin rights to access:

<https://github.com/uob-team-project/team00/settings/actions/runners/new>
team00 - replace with your team number



Adding a self-hosted runner requires that you download, configure, and execute the GitHub Actions Runner. By downloading and configuring the GitHub Actions Runner, you agree to the [GitHub Terms of Service](#) or [GitHub Corporate Terms of Service](#), as applicable.

Runner image



Architecture

x64

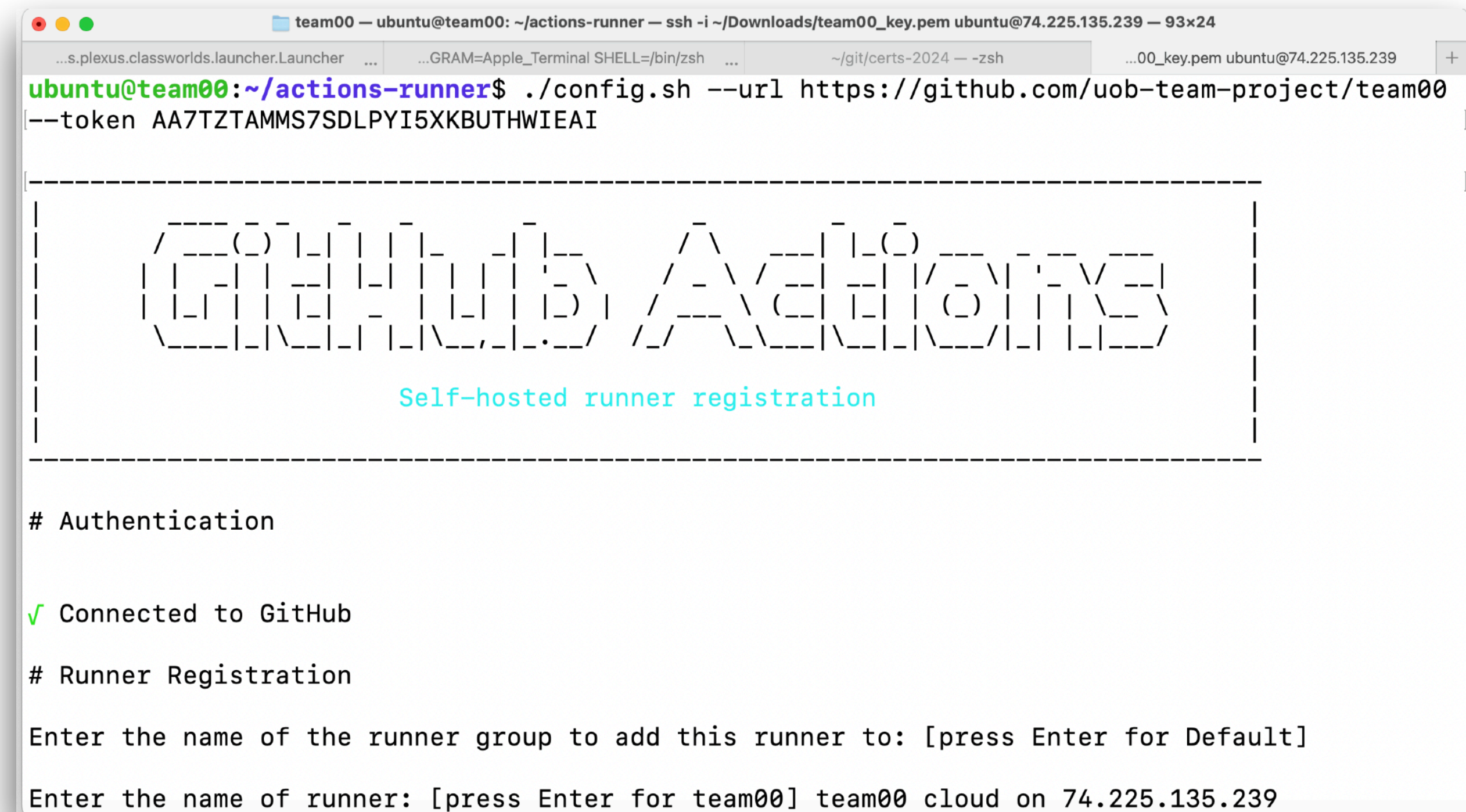
Download

```
# Create a folder
$ mkdir actions-runner && cd actions-runner

# Download the latest runner package
$ curl -o actions-runner-linux-x64-2.322.0.tar.gz -L https://github.com/actions/runner/releases/download/v2.322.0/actions-runner-linux-x64-2.322.0.tar.gz

# Optional: Validate the hash
$ echo "b13b784808359f31bc79b08a191f5f83757852957dd8fe3dbfcc38202ccf5768  actions-runner-linux-x64-2.322.0.tar.gz" | shasum -a 256 -c

# Extract the installer
$ tar xzf ./actions-runner-linux-x64-2.322.0.tar.gz
```



Instead of ./run do:

```
sudo ./svc.sh install
sudo ./svc.sh start
```

Add GitHub secrets



https://github.com/uob-team-project/team00/settings/secrets/actions 90% ☆ ⬇

uob-team-project / team00

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General

Access

Collaborators and teams

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Rules

Actions

Webhooks

Environments

Codespaces

Pages

Custom properties

Security

Code security

Deploy keys

Secrets and variables

Actions

Codespaces

Actions secrets and variables

Secrets and variables allow you to manage reusable configuration data. Secrets are **encrypted** and are used for sensitive data. [Learn more about encrypted secrets](#). Variables are shown as plain text and are used for non-sensitive data. [Learn more about variables](#).

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.

Secrets Variables

Environment secrets

This environment has no secrets.

Manage environment secrets

Repository secrets

This repository has no secrets.

New repository secret

Organization secrets

Actions secrets / New secret

Name *

DEPLOY_USER

Secret *

ubuntu

Actions secrets / New secret

Name *

RSA_PRIVATE_KEY

Secret *

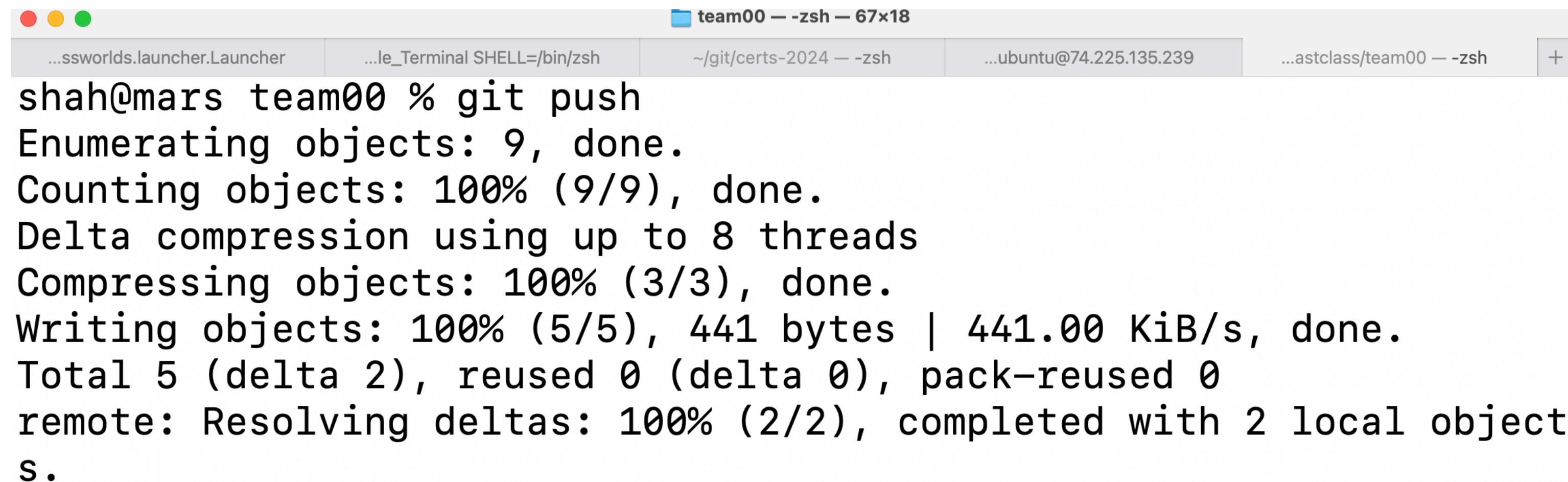
Lhm7ITDPpIT9ZDVMAnpMDu99D5hCxjqYuVlit8dw5G8drdC
5cGV8VvxeQIyy42vrmaOwwxVTPRjxDSQ/R8kjSLJy3iPafPBf
PE9rR8v0GD9q48AVYIDk23ewVIUvOFZBBcv1U+IbNwgJzU
7PAQdrguZlqdl+7yQwxvFMuyh+MbpT2oTQ/P8EQMRM+78H
q4s2bl5YwIVptWviHDZ48/eLYW3zCdZ5TreudbOWFHgYTiK

Repository secrets

Name
DEPLOY_IP
DEPLOY_USER
RSA_PRIVATE_KEY

**commit to main so jobs
publish-docker: and
deploy-dev: will run**

```
git commit -m "development deployment"  
git push
```

A terminal window titled "team00 — -zsh — 67x18" with a tab bar showing several open tabs. The active tab is "team00 — -zsh — 67x18". The terminal output shows the execution of "git push" and its progress.

```
shah@mars team00 % git push  
Enumerating objects: 9, done.  
Counting objects: 100% (9/9), done.  
Delta compression using up to 8 threads  
Compressing objects: 100% (3/3), done.  
Writing objects: 100% (5/5), 441 bytes | 441.00 KiB/s, done.  
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0  
remote: Resolving deltas: 100% (2/2), completed with 2 local object  
s.
```

Check the workflows ran without errors

The screenshot displays the GitHub Actions interface for the repository 'uob-team-project / team00'. The main view shows 'All workflows' with a list of 16 workflow runs. The first three runs are for the workflow 'build and deploy jhipster application', all with a status of 'Success'. The third run is highlighted, showing its details.

Workflow Run Details:

- Workflow:** build and deploy jhipster application
- Run #16:** build and deploy jhipster application #16
- Status:** Success
- Triggered via push:** 34 minutes ago
- Total duration:** 3m 58s
- Artifacts:** -

Jobs:

- ✓ Publish Docker
- ✓ deploy development app via ssh
- ⌚ deploy production version via ssh

Run details:

- Usage

main.yml:

```
on: push
```

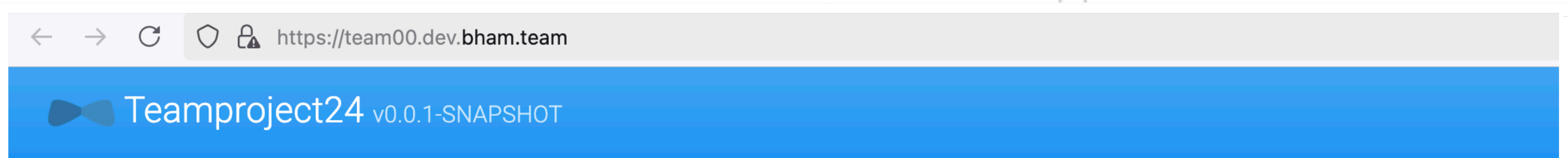
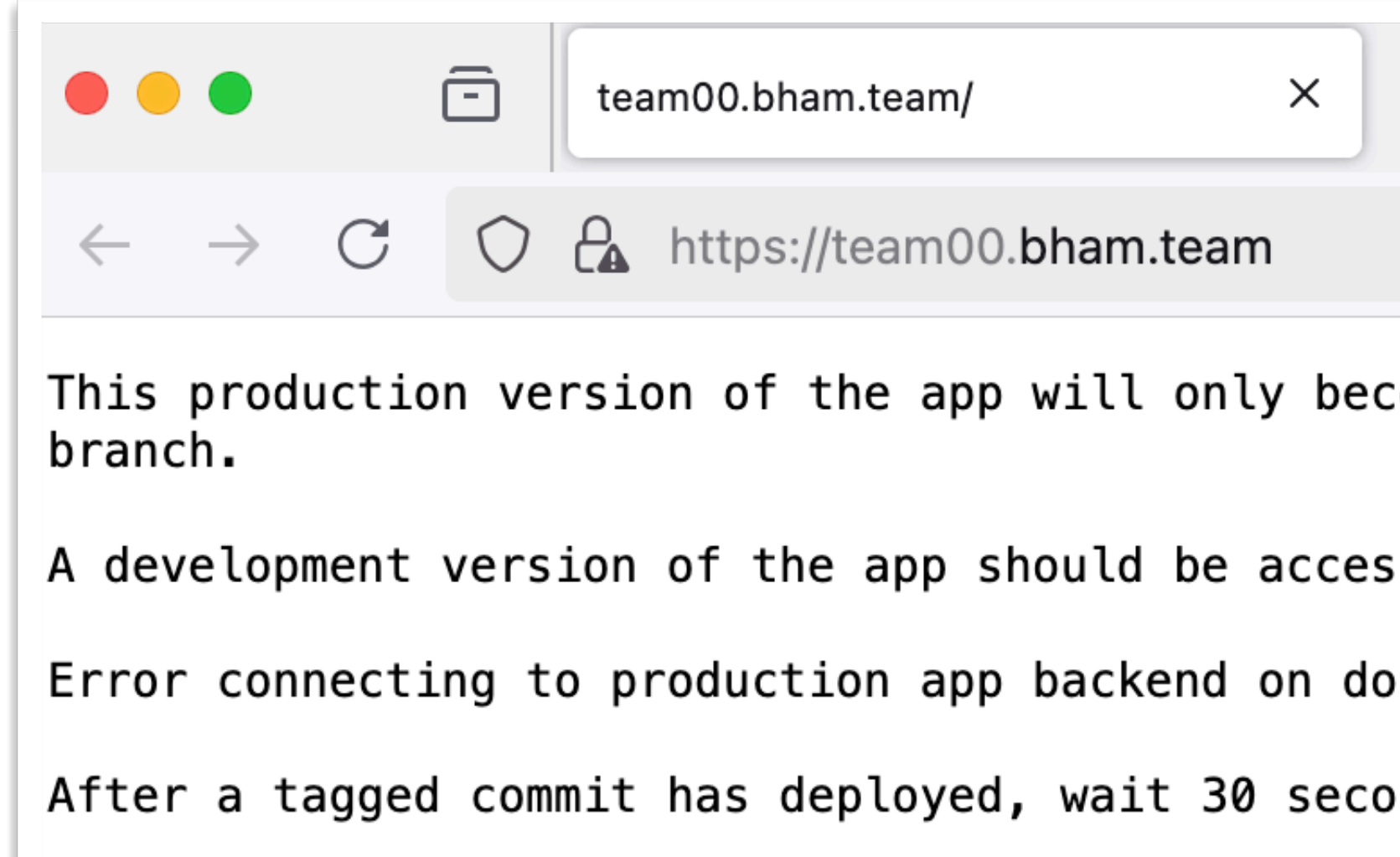
Workflow Progress:

- ✓ Publish Docker (3m 23s)
- ✓ deploy development app v... (18s)

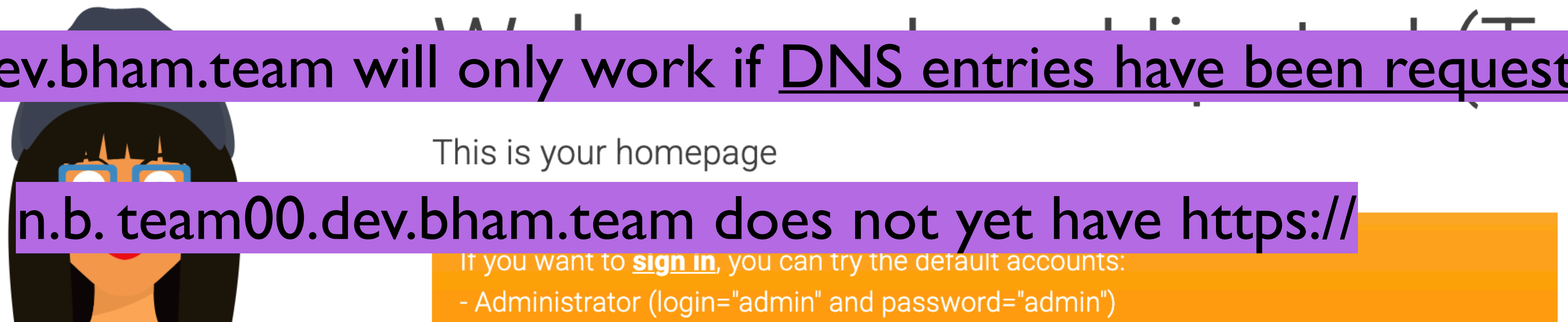
Check team00.dev.bham.team is up

Caddy web server is running on 74.225.135.239

After getting DNS entries from <https://forms.office.com/e/C0kaiRMuDD>
try: <http://team00.dev.bham.team> or
<http://team00.bham.team>



n.b. team00.dev.bham.team will only work if DNS entries have been requested



n.b. team00.dev.bham.team does not yet have https://

Create a tagged commit so job deploy-prod: will also run

```
git tag -a v0.1 -m "first tagged commit so job publish-docker runs"  
git push origin --tags
```

```
shah@mars team00 % git tag -a v0.4 -m "fourth tagged commit"  
shah@mars team00 % git push origin --tags  
Enumerating objects: 1, done.  
Counting objects: 100% (1/1), done.  
Writing objects: 100% (1/1), 162 bytes | 162.00 KiB/s, done.  
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0  
To github.com:uob-team-project/team00.git  
* [new tag]          v0.4 -> v0.4  
shah@mars team00 %
```

Check the workflow `deploy-prod`: ran without errors

- build and deploy jhipster application

✓ **lecture #40**

Re-run

Summary

Jobs

✓ Publish Docker

✓ deploy development app via ssh

✓ deploy production version via ssh

Run details

Usage

Workflow file

Triggered via push 50 minutes ago

madasar pushed  b5a8d86 `v0.21`

Status

Success

Total duration

4m 28s

Artifacts

—

main.yml

on: push

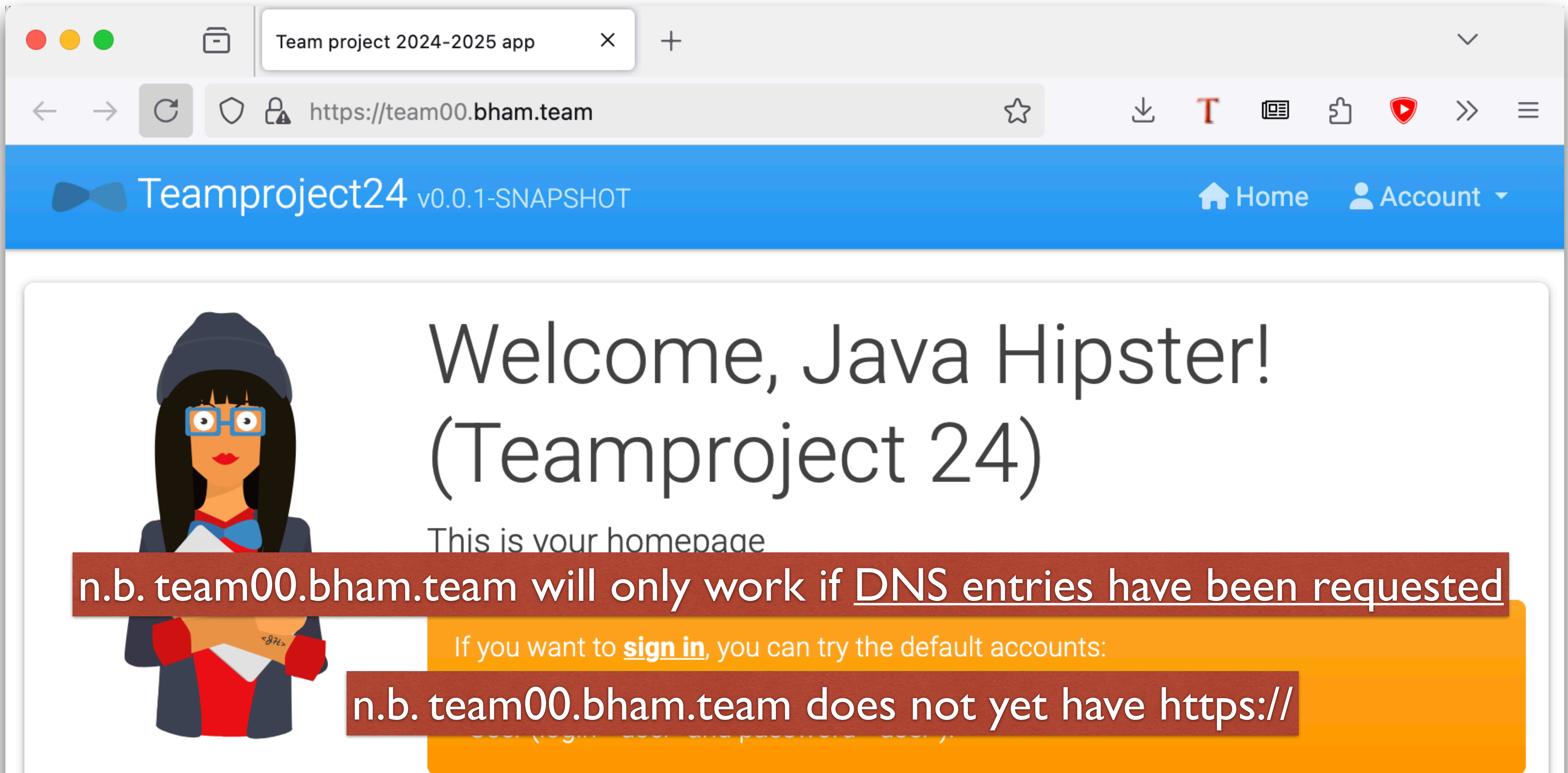
✓ Publish Docker

3m 30s

✓ deploy development app v... 18s

✓ deploy production version ... 11s

Check team00.bham.team is up



Team project 2024-2025 app

https://team00.bham.team

Teamproject24 v0.0.1-SNAPSHOT

Home Account

Welcome, Java Hipster!
(Teamproject 24)

This is your homepage

n.b. team00.bham.team will only work if DNS entries have been requested

If you want to sign in, you can try the default accounts:

n.b. team00.bham.team does not yet have https://

Enable https:// in `.github/workflows/main.yml`

n.b. only enable https:// if DNS entries exist and
team00.dev.bham.team / team00.bham.team
can be opened in a web browser

Change line 129:

```
ACME=${{ env.ACME_STAGING }} \
```

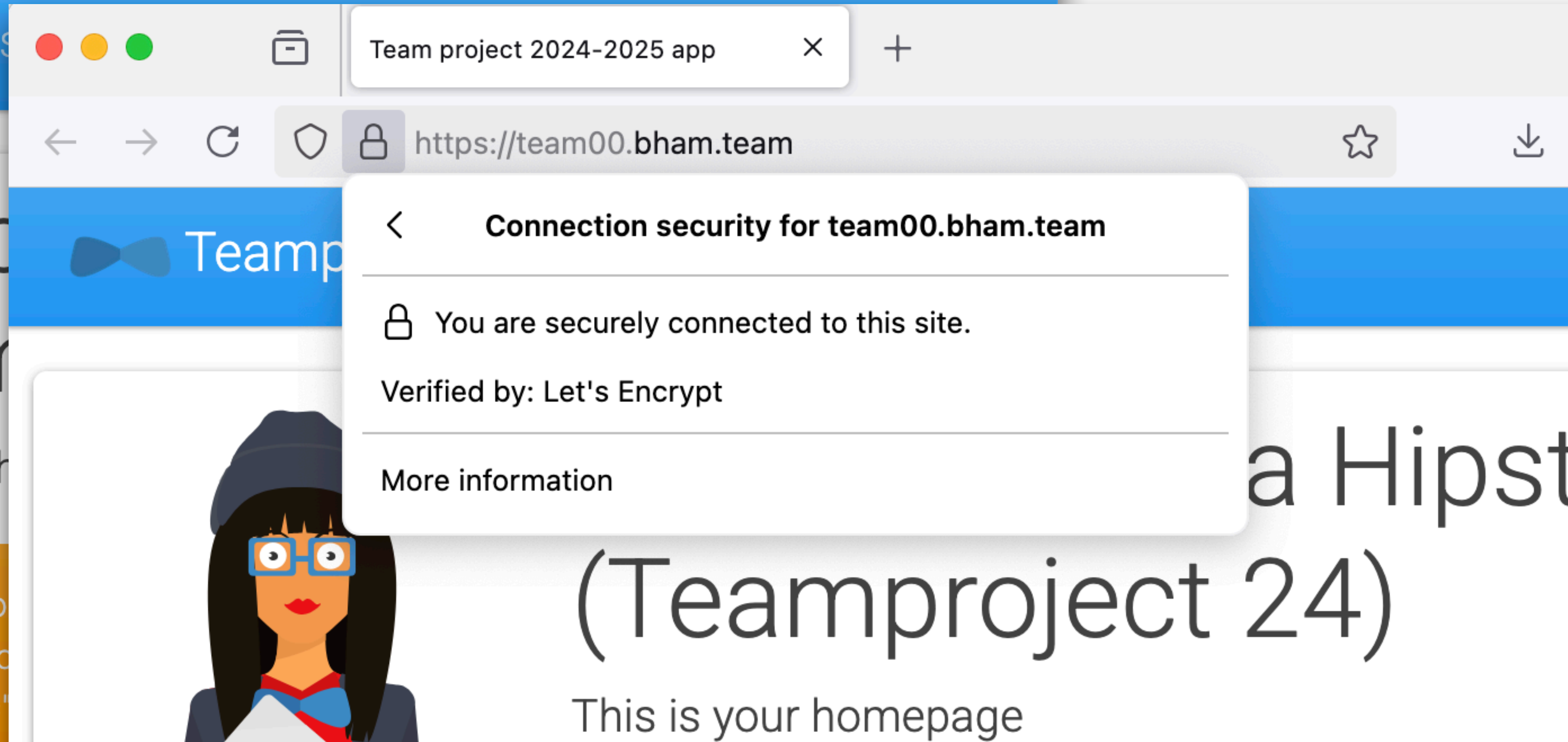
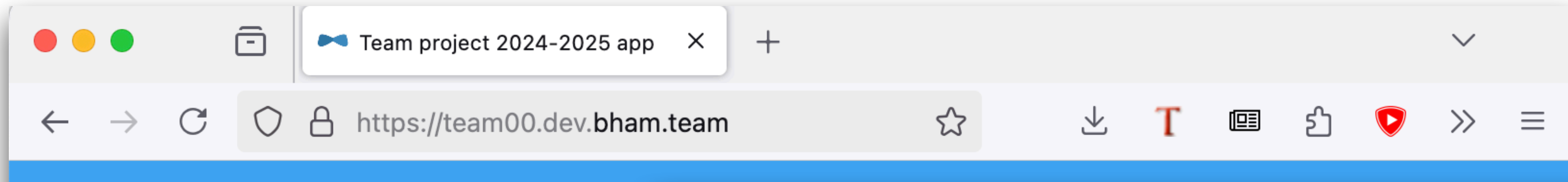
To:

```
ACME=${{ env.ACME }} \
```

Git commit to main branch to get into production:

```
git commit -m "development deployment"  
git push
```

Check team00.bham.team and
team00.dev.bham.team have https:// 



Get Custom URL working: edit Caddyfile

```
#to serve on a CUSTOMURL e.g. skeletons.bham.team
# 1) request a DNS entry for skeletons.bham.team pointing at the Virtual
Machine IP address (4.213.95.197) https://forms.office.com/e/C0kaiRMuDD
# 2) replace '#CUSTOMURL' with 'skeletons.bham.team', uncomment the last
3 lines of the file (lines 49-51)
Caddyfile
# 3) make a tagged commit
#CUSTOMURL {
#   handle_errors {
#       respond "CUSTOMURL not working"
#   }
#   reverse_proxy http://teamproject-prd:8080
#}
```

```
#to serve on a CUSTOMURL e.g. skeletons.bham.team
# 1) request a DNS entry for skeletons.bham.team pointing at the Virtual
Machine IP address (4.213.95.197) https://forms.office.com/e/C0kaiRMuDD
# 2) replace '#CUSTOMURL' with 'skeletons.bham.team', uncomment the last
lines of the file (lines 49-54) in the git repo file src/main/docker/Caddyfile
# 3) make a tagged commit
skeletons.bham.team {
    handle_errors {
        respond "CUSTOMURL not working check team00.bham.team is up"
    }
    reverse_proxy http://teamproject-prd:8080
}
```

Git commit to main branch to get into production:

```
git commit -m "development deployment"
git push
```

Custom URL: edit Caddy file

src/main/docker/Caddyfile

Follow the instructions at the bottom of the file:

```
#to serve on a CUSTOMURL e.g. skeletons.bham.team
# 1) request a DNS entry for skeletons.bham.team pointing at
# the Virtual Machine IP address (DEPLOY_IP) https://forms.office.com/e/C0kaiRMuDD
# 2) replace '#CUSTOMURL' with 'skeletons.bham.team', uncomment the last lines of the file
# (lines 49-54) in the git repo file src/main/docker/Caddyfile
# 3) make a tagged commit
#CUSTOMURL {
#  handle_errors {
#    respond "CUSTOMURL not working check DOMAIN is up"
#  }
#  reverse_proxy http://teamproject-prd:8080
#}
```

Check CUSTOMURL is up and has https://



← → ↻ 🔒 https://skeletons.bham.team ☆

< **Connection security for skeletons.bham.team**

🔒 You are securely connected to this site.

Verified by: Let's Encrypt

More information

Teamp

Welcome, Java Hipster! (Teamp

This is your homepage

If you want to **sign in**, you can try the default accounts:

- Administrator (login="admin" and password="admin")
- User (login="user" and password="user").

You don't have an account yet? **[Register a new account](#)**

If you have any question on JHipster:



Next steps:

Develop MVP - generate app from JDL,

1. pick up ticket
2. commit to main
3. get feedback from team and end users
4. tagged commit
5. repeat 1)-4) until M2/M3 submission :)

Agile Team Project

Week 5: Part 3

Debugging tips and commands for deployment

get latest version of the CI files using the terminal

- 1) cd to the root folder of your repo
- 2) Run the following commands:



```
curl -fsSL https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/.github/workflows/main.yml -o .github/workflows/main.yml
curl -fsSL https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/Caddyfile -o src/main/docker/Caddyfile
curl -fsSL https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/dev.yml -o src/main/docker/dev.yml
curl -fsSL https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/prd.yml -o src/main/docker/prd.yml
curl -fsSL https://raw.githubusercontent.com/uob-team-project/team00/refs/heads/main/src/main/docker/install-app.sh -o src/main/docker/install-app.sh
```

<https://code-pad.me/winter-oklahoma-table-connecticut>

After this: commit to git

Build the app locally using docker

Find any docker build issues with the app

ARM64/M1 Mac:

```
npm run java:docker:arm64
```

```
[INFO] Using base image with digest: sha256:08414944e310
5ac27b1bf31bd3878bc1f1
[INFO]
[INFO] Container entrypoint set to [bash, -c, /entrypoint
[INFO]
[INFO] Built image to Docker daemon as teamproject24
[INFO] Executing tasks:
[INFO] [=====] 100.0% complete
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 22.457 s
[INFO] Finished at: 2025-02-20T12:10:40Z
[INFO] -----
```

AMD64/Intel:

```
npm run java:docker
```

Make sure docker is installed and Docker desktop is running

Run the app locally using docker compose

To start the app: Find any docker runtime issues with the app

```
docker compose -f src/main/docker/app.yml up
```

```
app-1          | 2025-02-20T12:14:40.073Z  INFO 1 --- [           main] org.jboss.tnreads
hreads version 3.5.0.Final
app-1          | 2025-02-20T12:14:40.105Z  INFO 1 --- [           main] bham.team.Teamproject24A
Teamproject24App in 7.081 seconds (process running for 7.414)
app-1          | 2025-02-20T12:14:40.108Z  INFO 1 --- [           main] bham.team.Teamproject24A
app-1          | -----
app-1          |           Application 'teamproject24' is running! Access URLs:
app-1          |           Local:           http://localhost:8080/
app-1          |           External:        http://172.18.0.3:8080/
app-1          |           Profile(s):      [prod, api-docs]
app-1          | -----
v View in Docker Desktop  o View Config  w Enable Watch
```

To take the app down:
Ctrl + c

Make sure the docker image has been built before docker compose is run
Stop postgres if you have installed before if it is running

confirm domain names are live using nslookup

nslookup can help confirm the DNS entries are live:

```
nslookup skeletons.bham.team
```

```
Server:      8.8.8.8
Address:     8.8.8.8#53
```

```
nslookup team00.bham.team
```

```
Non-authoritative answer:
Name:   skeletons.bham.team
Address: 4.213.95.197
```

```
nslookup team00.dev.bham.team
```

```
skeletons.team00 % █
```

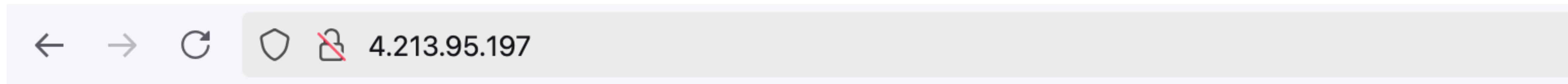
```
Server:      8.8.8.8
Address:     8.8.8.8#53
```

```
** server can't find notlive.bham.team: NXDOMAIN
```

```
skeletons.team00 % █
```

confirm caddy server is live after deployment

Using a web browser we can confirm some part of the server is running



Caddy web server is running on 4.213.95.197

After getting DNS entries from <https://forms.office.com/e/C0kaiRMuDD>
try: <http://team00.dev.bham.team> or
<http://team00.bham.team>

ssh into the cloud virtual machine

to try and debug issues with deployment

```
chmod go-rx ~/Downloads/team99_key.pem
```

```
[shah@mars mvp % ls -la ~/Downloads/team99_key.pem  
-rw-----@ 1 shah  staff  2494 20 Feb 11:39 /Users/shah/Downloads/team99_key.pem
```

```
ssh -i ~/Downloads/team99_key.pem ubuntu@98.70.112.45
```

Or

```
ssh -i ~/Downloads/team99_key.pem ubuntu@team99.bham.team
```

The key must be only readable by your user (chmod + ls -la to confirm)

To do **ssh ... team99.bham.team** : must have a DNS entry pointing to VM

See or edit files on cloud VM

cat to display file contents, **nano** to edit

cat filename

nano filename

e.g.:

cat team-project/dev.yml

cat team-project/prd.yml

nano caddy/Caddyfile

Cloud vm file structure

To verify the files are all there

```
[ubuntu@team99:~$ tree team-project/
```

```
team-project/
├── Caddyfile
├── app.yml
├── caddy.yml
├── dev.yml
├── grafana
│   ├── provisioning
│   │   ├── dashboards
│   │   │   ├── JVM.json
│   │   │   └── dashboard.yml
│   │   └── datasources
│   │       └── datasource.yml
├── install-app.sh
├── install-docker.sh
├── jhipster-control-center.yml
├── jib
│   └── entrypoint.sh
├── monitoring.yml
├── postgresql.yml
├── prd.yml
├── prometheus
│   └── prometheus.yml
├── services.yml
└── sonar.yml
```

7 directories, 17 files

```
[ubuntu@team99:~$ ls caddy
```

```
Caddyfile  config  data
```

```
[ubuntu@team99:~$ ls team-project/
```

```
Caddyfile  dev.yml  install-docker.sh  monitoring.yml  prometheus
app.yml    grafana  jhipster-control-center.yml  postgresql.yml  services.yml
caddy.yml  install-app.sh  jib  prd.yml  sonar.yml
```

```
[ubuntu@team99:~$ ls
```

```
actions-runner  caddy  get-docker.sh  prd  prd.current.yml  team-project
```

See what is running on docker

to check that the app (prd and dev and database) were able to start up

docker ps

```
[ubuntu@team99:~]$ docker ps
```

CONTAINER ID	IMAGE	COMMAND
CREATED	STATUS	PORTS
NAMES		
92ed84b6ae0c	docker:dind	"dockerd-entr
point...."	2 minutes ago	Up 2 minutes
0.0.0.0:2375->2375/tcp, [::]:2375->2375/tcp, 2376/tcp		cde1058fd2b44ce6a79b
8df5c3d0c10_dockerdind_2a9bac		
0bfad9f9ffef	ghcr.io/uob-team-project/uob-team-project/team99:ba76c246e03e57fef8cc94c584bc5c1b4452942f	"bash -c /ent
ypoint..."	34 minutes ago	Up 34 minutes
0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp		team-project-teampro
ect-prd-1		
07f4d8356043	postgres:14.5	"docker-entry
oint.s..."	34 minutes ago	Up 34 minutes
5432/tcp		team-project-teampro
ect-postgresql-prd-1		
8266541b25fb	postgres:14.5	"docker-entry
oint.s..."	34 minutes ago	Up 34 minutes
5432/tcp		team-project-teampro
ect-postgresql-1		
3d85beaaea2d	ghcr.io/uob-team-project/uob-team-project/team99:latest	"bash -c /ent
ypoint..."	34 minutes ago	Up 34 minutes
0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp		team-project-teampro
ect-dev-1		

Cloud vm: start and stop development app

Take the development app down

```
docker compose -f team-project/dev.yml down
```

Take the development app up **with debug messages**

```
docker compose -f team-project/dev.yml up
```

Take the development app up **in background no debug**

```
docker compose -f team-project/dev.yml up -d
```

debug/error messages will help understand and fix errors

Cloud vm: start and stop production app

Take the production app down

```
docker compose -f team-project/prd.yml down
```

Take the production app up **with debug messages**

```
docker compose -f team-project/prd.yml up
```

Take the production app up **in background no debug**

```
docker compose -f team-project/prd.yml up -d
```

debug/error messages will help understand and fix errors

Cloud vm: start and stop caddy web server

Take caddy down

```
docker compose -f team-project/caddy.yml down
```

Take caddy up **with debug messages**

```
docker compose -f team-project/caddy.yml up
```

Take caddy up **in background no debug**

```
docker compose -f team-project/caddy.yml up -d
```

debug/error messages will help understand and fix errors

Reset the production database on cloud VM

if the database schema/JDL changes it can stop the production app working

1. **ssh in; stop the production app:** `docker compose -f team-project/prd.yml down`

2. **Either** if the database is not valuable *remove* the prd database folder:

`sudo rm -rf prd`

3. **or** *move* the prd database folder to a backup e.g. prd-backup-date:

`sudo mv prd prd-backup-21-02-25`

4. **Start the production app:** `docker compose -f team-project/prd.yml up -d`

5. n.b. development app database is reset every deploy (not saved)

Recreate the main branch from template team00 repo

Useful if JDL needs to change or the app breaks

1. `git checkout main ; echo switch to the main branch`
2. `git rm -r '*' ; echo remove all the files in the main branch`
3. `git commit -m "removed old main branch"`
4. `curl -L https://github.com/uob-team-project/team00/archive/refs/heads/main.zip -o team00.zip ; echo download the team00.zip file with the template repo`
5. `unzip team00.zip; mv team00-main/. * .; mv team00-main/* . ; echo extract team.zip here`
6. `rm -r team00-main team00.zip ; echo remove the zip file and team00-main folder`
7. `./mvnw ; echo confirm the template app works`
8. `git add . ; echo add the new template app to git`
9. `git commit -m "added template app back to repository"`
10. `git push ; echo confirm the app deployed and works`
11. optional: re generate from JDL
12. optional: copy back work from the git history using `git cherrypick` or `git checkout file:hash`

Re generate the app with different JHipster settings

Useful if JHipster settings e.g. oauth/websockets/frontend/theme need to change

1. `git checkout main ; echo` switch to the main branch of the template app
2. `git rm -r '*' ; echo` remove all the files in the main branch
3. `git commit -m "removed old main branch"`
4. `jhipster ; echo` regenerate the app with the new settings
5. download the files from the template repo customised by madasar for CI/footer/GDPR.
use the curl commands on this link, run in the root of the repo:

<https://code-pad.me/xray-alaska-seven-yellow>

(the gdpr-policy and footer may not make sense if the front end is changed to react/vue)
6. `./mvnw ; echo` confirm the app works
7. `git add .`
8. `git commit -m "added template app back to repository"`
9. `git push ; echo` confirm the app deployed and works
10. optional: re generate from JDL
11. optional: copy back work from the git history using `git cherrypick` or `git checkout file:hash`